Lab 13 — Windows Containers Fundamentals (Docker Desktop & Windows Server)

K8S-LAB-13 · Windows · Beginner · 60 min

Focus: Windows containers, Docker Desktop, process isolation, Hyper-V isolation, ltsc2022

Objective

Run your first Windows container on Windows Server 2022 / Docker Desktop, understand process vs Hyper-V isolation, and inspect the Windows container runtime.

Concepts Covered

• Process isolation vs Hyper-V isolation

• Why Windows host build must match container base image (ltsc2019 vs ltsc2022)

• nanoserver vs servercore base images

• Docker Desktop Windows-containers mode vs Linux-containers mode

Prerequisites

• Windows 10/11 Pro or Windows Server 2022

• Docker Desktop installed

• Virtualization enabled in BIOS

Architecture

```
Windows Host (Server 2022 / Win11)
  |
  +-- Docker Engine (Windows mode)
        |
        +-- Process-isolated container  (shares host kernel)
        |
        +-- Hyper-V isolated container  (tiny utility VM)
```

Lab Steps

01. Switch Docker Desktop to Windows containers

```
& 'C:\Program Files\Docker\Docker\DockerCli.exe' -SwitchDaemon
docker version
```

Expected: Server: OS/Arch: windows/amd64 in docker version output.

02. Pull and run nanoserver

```
docker pull mcr.microsoft.com/windows/nanoserver:ltsc2022
docker run --rm mcr.microsoft.com/windows/nanoserver:ltsc2022 cmd /c "ver && hostname"
```

Expected: Microsoft Windows version + container hostname.

03. Run with Hyper-V isolation

```
docker run --rm --isolation=hyperv mcr.microsoft.com/windows/nanoserver:ltsc2022 cmd /c "ver"
```

Expected: Same output, ~3-5s slower to start.

04. Inspect the running engine

```
docker info | findstr /i "isolation osversion"
Get-Service docker
```

Expected: Default Isolation: process, OSVersion: 10.0.20348.x

Common Errors

hcsshim CreateComputeSystem: OS does not match
Cause: Image base build differs from host build
Fix: Use --isolation=hyperv OR pull matching ltsc tag

Docker Desktop hangs on switch
Cause: Hyper-V/Containers feature missing
Fix: Enable-WindowsOptionalFeature -Online -FeatureName Containers,Microsoft-Hyper-V -All

Troubleshooting

• docker events

• Get-WinEvent -LogName Microsoft-Windows-Hyper-V-Compute-Admin

• docker system info

Enterprise Relevance

Windows containers underpin all .NET Framework modernization. Knowing isolation modes lets you decide between density (process) and tenant isolation (hyperv).

Validation

• docker version shows windows/amd64 server

• Both process and hyperv containers run on the same host

Questions

[Beginner] Q: Can a Windows container run a Linux binary?

A: No — the kernel ABI is different.

[Intermediate] Q: When do you choose Hyper-V isolation?

A: Multi-tenant hosts, untrusted images, mismatched builds.

[Advanced] Q: Why prefer nanoserver over servercore?

A: Smaller attack surface, ~10x smaller image, faster pulls.

[Scenario] Q: docker run fails with 'OS does not match'. Fix?

A: Add --isolation=hyperv or pull the matching ltsc tag.

[Viva] Q: Docker Desktop Linux vs Windows mode?

A: Linux uses WSL2 VM; Windows talks to native dockerd.exe.